

Gold

79
Au The chemical symbol for this element comes from the Latin word *aurum*, meaning “gold.” This metal is found in its pure form and has been used in jewelry and decoration for more than 5,000 years. Alchemists—the forerunners of modern chemists—sought to convert relatively common metals such as lead into precious ones like gold.

Reflecting heat

The sun's heat can be dangerous to equipment or astronauts in space.

As a safety measure, an astronaut wears a helmet with a visor covered with a thin layer of gold.

This protective coating reflects the sun's intense, harmful rays away from the astronaut's face and prevents overheating.



Gold used as a reflective screen



Flakes of pure gold refined in a laboratory

Gold in quartz

Gold is mostly found as veins in rocks such as quartz. These form when a hot mixture of minerals rises from deep within Earth in liquid form, then cools and solidifies near the surface.

When these rocks are broken down, by weathering or river erosion, particles of gold are released.

Coins of gold

Pure gold and alloys that include gold have been used in coins since ancient times. Its shiny appearance, high value-to-weight ratio, and ability to resist corrosion made it a popular choice for use as money.



Coin issued by King Philip II of Macedonia (359–336 BCE)

Decorating with gold

Gold does not tarnish over time and as such has been used in buildings by architects since ancient times as an adornment. Gold leaf work and gold plating are common forms of decorations. The metal can even be used in flake and dust form to decorate food for consumption in some parts of the world, such as India.



The gold covering the Golden Temple in Amritsar, India, is not only decorative; it also resists corrosion.